

RCIM Model-Bank Reproduction MLP Residual Cell Final Closure Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the dedicated exact-paper residual-cell MLP wave prepared in:

- `doc/reports/campaign_plans/track_1/exact_paper/2026-04-21-23-32-36_track1_mlp_residual_cell_final_closure_campaign_plan_report.md`

The campaign targeted only the last four distinct accepted MLP family-target pairs that were still non-green in the canonical RCIM Model-Bank Reproduction full-matrix benchmark.

- campaign name: `track1_mlp_residual_cell_final_closure_campaign_2026_04_21_23_32_36`
- report timestamp: `2026-04-22-01-08-33`
- completed validation runs: `216`
- failed validation runs: `0`
- promoted targeted pairs: `4/4`
- retained baseline pairs: `0/4`
- visibly promoted benchmark cells: `7/8`
- campaign artifact root:
`output/training_campaigns/track1/exact_paper/forward/family_repair/mlp/track1_mlp_residual_cell_final_closure_campaign_2026_04_21_23_32_36`

Campaign Winner

- Run: `track1_mlp_amplitude_156_final_closure_attempt_32`
- Scope: `amplitudes_only`
- Closure Score: `1.000`
- Met / Near / Open: `2 / 0 / 0`

Targeted Pair Outcome

The table below compares the accepted pre-wave baseline against the best campaign-local candidate and the final accepted post-wave value for each targeted residual MLP pair. Each metric cell is shown as MAE / RMSE .

Pair	Baseline	Campaign Best	Accepted	Source	Result
A1	0.00755033 / 0.0137694	0.00476074 / 0.00898767	0.00476074 / 0.00898767	campaign	fully_green
A156	0.00682945 / 0.011491	0.00644413 / 0.0106664	0.00644413 / 0.0106664	campaign	fully_green
A240	0.00638307 / 0.0121452	0.00524356 / 0.00911022	0.00524356 / 0.00911022	campaign	fully_green
phi162	0.826864 / 1.33334	0.826864 / 1.29701	0.826864 / 1.29701	mixed	still_non_green

Accepted MLP Row After Closeout

- Table 2 remaining non-green harmonics: none
- Table 3 remaining non-green harmonics: none
- Table 4 remaining non-green harmonics: 162
- Table 5 remaining non-green harmonics: 162
- total remaining non-green MLP cells on Tables 2-5 : 2

Canonical RCIM Model-Bank Reproduction Impact

- This residual MLP wave updates the accepted MLP family row without changing the canonical RCIM Model-Bank Reproduction scope definition.
- The benchmark row now keeps the better visible metric cell between the accepted baseline and this dedicated residual closure wave.
- The cross-family closure counts should now be read directly from the refreshed benchmark after this closeout.

Resulting Canonical State

- benchmark path: doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md
- master summary path: doc/reports/analysis/Training Results Master Summary.md
- final closeout report path: doc/reports/campaign_results/track_1/exact_paper/forward/2026-04-22-01-08-33_track1_mlp_residual_cell_final_closure_campaign_results_report.md